Shi Quan, FOO

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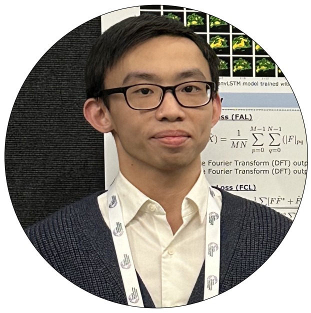

Machine Learning researcher specializing in generative and spatiotemporal models. Co-author of NeurIPS 2024 and TMLR 2025 papers on precipitation nowcasting. Experienced in building PyTorch models and processing large-scale radar datasets.

EDUCATION

Bachelor of Science with Major in Physics — Sep 2018 - Aug 2022

The Hong Kong University of Science and Technology (HKUST), Hong Kong — GPA: 3.811/4.300

PUBLICATIONS

STLDM: Spatio-Temporal Latent Diffusion Model for Precipitation Nowcasting. Transactions on Machine Learning Research (TMLR), 2025. [Paper] [Code] [Demo]

Fourier Amplitude and Correlation Loss: Beyond Using L2 loss for Skillful Precipitation Nowcasting. NeurIPS 2024. [Paper] [Code]

PROFESSIONAL AND RESEARCH EXPERIENCE

Research Assistant | *Prof. Yeung Dit-Yan's lab, HKUST, Hong Kong* — Aug 2022 - Present

- Designed large-scale data pipelines processing radar reflectivity and rainfall datasets for spatiotemporal forecasting models.
- Developed **FACL**, a Fourier-based loss that improves precipitation nowcasting visual sharpness by 50%. This work was accepted at NeurIPS 2024.
- Developed **STLDM**, a spatiotemporal latent diffusion model for precipitation nowcasting, achieving SOTA performance across multiple radar datasets and 10× faster inference. This work was accepted at TMLR 2025.
- Released official open-source implementation and deployed an interactive Gradio demo for model inference.

Summer Intern | *Hong Kong Observatory (HKO), Hong Kong* — Jun 2021 - Jul 2021

- Analyzed historical rainfall time series data to estimate heavy rainfall probability using analog methods.
- Cleaned and aggregated time-series rainfall data to support statistical analysis.
- Presented statistical rainfall analysis results to the meteorological staff.

Undergraduate Researcher | *Prof. Jo Gyu-Boong's Lab, HKUST, Hong Kong* — Feb 2021 - Dec 2021

- Developed software to control and calibrate the Digital Micro-Mirror Device (DMD) for laser beam modulation.
- Collected and validated experimental optical data, ensuring the measurement consistency.

Undergraduate Researcher | *Prof. Du Shengwang's Lab, HKUST, Hong Kong* — Apr 2019 - Jun 2020

- Built a PID-controlled circuit for laser intensity stabilization.
- Simulated and conducted a theoretical study on an Electromagnetically Induced Transparency (EIT) system.
- Operated Fabry-Pérot filter to isolate specific wavelength ranges, gaining hands-on experience with optical instruments.

TECHNICAL SKILLS

- **Programming:** Python
- **ML Frameworks:** PyTorch, TensorFlow
- **ML Techniques:** Generative Models, Diffusion Models, Spatiotemporal Modeling, Time Series Forecasting
- **Tools:** Git, Linux, TensorBoard, Gradio, HuggingFace

HONORS & AWARDS

Paul Ching Wu Chu Scholarship for Physics Students	2021-2022
Physics Entry Scholarship	2019-2022
HKSAR Government Scholarship Fund - Belt and Road Scholarship (Malaysia)	2018-2022
Honorable Mention in 19th Asian Physics Olympiad	2018

REFERENCES

- **Prof. Yeung Dit-Yan**
  Chair Professor, Department of Computer Science and Engineering, HKUST — Available Upon Request